Week 12: 16th – 20th November 2026

Subject: Mathematics Pre-AP Geometry with Statistics **Unit Title: Unit 1 Performance Task / Unit 2: Tools and Techniques of Geometric Measurement** **Grade: 9**

Lessons	Learning targets	Assessment tools and strategies Formative/ Summative/ Differentiation	Resources and links for Parents
Lesson 1: Performance Task – Part D: Data Analysis'	I can analyse the data collected for my study by selecting and calculating at least three appropriate statistics, creating a suitable data display, and interpreting the results in relation to my research question.	Support—provide a data-type decision guide, worked example, calculation scaffold and display template; Core—complete analysis independently as a pair and explain results; Extension—compare alternative displays/statistics, evaluate limitations and strengthen the interpretation with additional evidence.	College Board Pre-AP Geometry with Statistics, Unit 2, Lesson 2.3 Student Resource, Handouts 2.3.A–2.3.C (pp. 11–17); graph paper, ruler, calculator/GDC, coordinate-grid slides and notebook.
Lesson 2: Performance Task – Part E: Conclusion and Presentation	I can answer my research question using evidence from my analysis and communicate my findings in a clear report and brief presentation.	Support—provide a report structure, presentation outline and conclusion sentence frames; Core—produce a complete evidence-based report and brief presentation; Extension—evaluate the strength and limitations of the evidence, qualify the conclusion appropriately and respond to challenging audience questions.	
Lesson 3: Lesson 2.2 – Copying Line Segments and Angles	I can construct a copy of a given line segment and angle using a straightedge and compass, and justify that the copies are congruent to the original figures.	Support—provide a labelled step sequence, demonstrate compass handling and offer partially marked diagrams; Core—complete segment and angle constructions and justify congruence; Extension—complete the equilateral-triangle/three-lamppost construction and explain why all three distances are equal.	
Lesson 4: Lesson 2.3 – Measuring Distance in the Coordinate Plane	I can calculate the distance between two points in the coordinate plane using the distance formula and apply it to solve geometric and real-world problems.	Support—provide a labelled coordinate grid, colour-coded horizontal/vertical changes and a formula scaffold; Core—solve distance problems and interpret results; Extension—solve reverse problems with an unknown coordinate and explain why more than one point may satisfy a fixed distance.	

Homework	Lesson 1: Complete the Part D analysis checklist. Finalise the required statistics, data display and interpretation; correct any calculation or display errors identified by the teacher. Lesson 2: Finalise the Part E report and presentation. Rehearse the presentation and ensure the conclusion is supported by evidence and acknowledges relevant limitations. Lesson 3: Complete selected practice from Handout 2.2.B. Review the compass-and-straightedge steps for copying a segment and an angle. Lesson 4: Complete selected questions from Handout 2.3.C. Show all working when applying the distance formula and correct errors from the exit ticket.
Teacher Notes	Students should bring notebooks, graphing calculators, rulers, compasses, straightedges, devices and all learning resources/handouts. Ensure access to their project data, the performance-task rubric, Student Resources 2.2 and 2.3, and lesson slides.

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Technology Integration	Microsoft Teams and OneNote; spreadsheet/graphing technology; GDC/calculators; performance-task rubric and project data; presentation software; straightedges, compasses, rulers, graph paper, coordinate grids and College Board Student Resources 2.2 and 2.3.
Cross-curricular links	Statistics and ICT: analysing real data, selecting summary statistics, constructing displays and communicating evidence through reports/presentations. Design and Technology: precision in straightedge-and-compass constructions. Geography/Urban Planning: scale maps, route distances and coordinate locations. Physics: distance as the length of a straight path and use of right-triangle relationships.
External Assessment Integration	College Board Pre-AP Geometry with Statistics, Unit 1 Performance Task: Designing a Study – Part D Data Analysis and Part E Conclusion and Presentation. Unit 2, Key Concept 2.1: Measurement in Geometry. Lesson 2.2: construct congruent copies of line segments and angles (Learning Objective 2.1.7). Lesson 2.3: calculate distance between points in the coordinate plane using the Pythagorean theorem (Learning Objective 2.1.8).
Wellbeing/ ATL Skills	Thinking Skills: select appropriate statistics, interpret distributions, evaluate evidence, construct geometric copies and apply the distance formula. Communication Skills: present findings clearly, justify conclusions, use precise construction language and show mathematical working. Self-Management: organise project tasks, check calculations and constructions, meet deadlines and prepare presentations. Social Skills: collaborate in project groups, peer-review reports and discuss construction strategies respectfully.
Standards	Unit 1 Performance Task – Designing a Study: Part D Data Analysis (summary statistics, data displays, symmetry/skew or independence, bias analysis); Part E Conclusion and Presentation (evidence-based conclusion, report and brief presentation). Unit 2: Tools and Techniques of Geometric Measurement, Key Concept 2.1. Learning Objective 2.1.7: Construct a congruent copy of a line segment or angle using straightedge and compass. Learning Objective 2.1.8: Calculate the distance between two points using the Pythagorean theorem in the coordinate plane.
Inquiry/ TOK Question	How can statistical evidence support a conclusion about a research question, and how can geometric constructions and the Pythagorean theorem help us copy figures and measure distances accurately?
Success Criteria (Measurable)	**Lesson 1: Performance Task – Part D: Data Analysis** • **All students must** calculate at least three appropriate statistics and include a suitable data display for their study. • **Most students should** interpret the statistics and analyse symmetry/skew or independence, as appropriate, in relation to the research question. • **Some students could** evaluate potential bias and limitations and explain how these affect the strength of the analysis.

	Lesson 2: Performance Task – Part E: Conclusion and Presentation • **All students must** write a conclusion that answers the research question and prepare a report of their findings. • **Most students should** support the conclusion with relevant statistics and communicate the results clearly in a brief presentation. • **Some students could** evaluate the strength of the evidence, qualify their conclusion and respond thoughtfully to questions. **Lesson 3: Lesson 2.2 – Copying Line Segments and Angles** • **All students must** use a straightedge and compass to construct a copy of a given segment and angle. • **Most students should** demonstrate accurate constructions and justify why the copied figures are congruent to the originals. • **Some students could** complete a multi-step construction, including an equilateral triangle, and explain the geometric reasoning. **Lesson 4: Lesson 2.3 – Measuring Distance in the Coordinate Plane** • **All students must** identify the horizontal and vertical changes between two plotted points and use them to calculate distance. • **Most students should** apply the distance formula accurately and interpret the result in a geometric or real-world context. • **Some students could** solve reverse-distance problems involving unknown coordinates and justify all possible solutions.
National Identity Domain	Respect: students give constructive feedback on peers' project reports and presentations, and value different valid construction strategies. Responsible decision-making: students support conclusions with appropriate evidence and use accurate geometric constructions and calculations.
	Area: Citizenship. Implementation: students present their findings clearly, consider the reliability of their data and acknowledge the limits of their conclusions.
	Area: Innovation and Responsible Decision-Making. Implementation: students use mathematical tools and digital resources to analyse evidence, construct geometric figures accurately and solve practical distance problems.